

Airline Cargo Demand Forecasting using EDA, Machine Learning and Deep Learning

Project Overview

Air cargo operations are a critical component of airline profitability. Efficient cargo planning helps airlines maximize aircraft utilization, reduce cargo offloading, improve customer satisfaction, and optimize revenue generation.

This project focuses on analyzing airline cargo operations data and forecasting future cargo demand using Exploratory Data Analysis (EDA), Machine Learning (ML), and Deep Learning (DL).

The objective is to identify operational patterns, measure key performance indicators (KPIs), predict cargo demand, and support data-driven decision-making for airline cargo operations.

Business Problem

Airlines often face challenges such as:

- Fluctuating cargo demand
- Aircraft capacity underutilization
- Cargo offloading
- Revenue leakage
- Flight delays impacting cargo movement
- Inefficient resource planning

Accurate demand forecasting can help airlines improve operational efficiency and profitability.

Dataset Information

Dataset Size

- 5,000 Records
- 11 Features

Features

Feature	Description
Date	Flight Date
Flight_No	Flight Number
Origin	Departure Station
Destination	Arrival Station
Aircraft_Type	Aircraft Type

Feature	Description
Capacity_KG	Available Cargo Capacity
Booked_KG	Cargo Booked
Loaded_KG	Cargo Loaded
Offloaded_KG	Cargo Offloaded
Revenue_INR	Revenue Generated
Delay_Minutes	Flight Delay

Target Variable

Booked_KG

Tools & Technologies

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- TensorFlow / Keras

Development Environment

- Jupyter Notebook

Version Control

- Git
 - GitHub
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Project Workflow

1. Data Collection

Historical airline cargo operations dataset.

2. Data Cleaning

- Missing Value Analysis
- Duplicate Detection
- Data Validation

3. Feature Engineering

Created operational KPIs:

- Load Factor
- Utilization
- Offload Percentage
- Revenue per KG

4. Exploratory Data Analysis

Performed:

- Demand Analysis
- Revenue Analysis
- Route Performance Analysis
- Aircraft Performance Analysis
- Delay Analysis
- Correlation Analysis

5. Machine Learning

Developed:

- Linear Regression
- Random Forest Regressor

Purpose:

Predict future cargo demand.

6. Deep Learning

Developed:

- LSTM Forecasting Model

Purpose:

Forecast future cargo demand trends.

Key Performance Indicators (KPIs)

Load Factor

$\text{Load Factor} = \text{Loaded Cargo} / \text{Capacity} \times 100$

Utilization

$\text{Utilization} = \text{Booked Cargo} / \text{Capacity} \times 100$

Offload Percentage

$\text{Offload \%} = \text{Offloaded Cargo} / \text{Booked Cargo} \times 100$

Revenue per KG

$\text{Revenue per KG} = \text{Revenue} / \text{Loaded Cargo}$

Exploratory Data Analysis

Analysis Performed

- ✓ Cargo Demand Distribution
- ✓ Revenue Distribution
- ✓ Aircraft Utilization Analysis
- ✓ Route Performance Analysis
- ✓ Delay Analysis
- ✓ Correlation Analysis

Visualizations

- Histograms
 - Boxplots
 - Scatterplots
 - Heatmaps
 - Route Performance Charts
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Machine Learning Models

Linear Regression

Purpose:

Baseline cargo demand prediction model.

Advantages

- Simple
- Easy Interpretation

Random Forest Regressor

Purpose:

Advanced cargo demand prediction.

Advantages

- Handles non-linear relationships
- High prediction accuracy
- Robust performance

Evaluation Metrics

- Mean Absolute Error (MAE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score
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Deep Learning Model

Long Short-Term Memory (LSTM)

Objective

Forecast future cargo demand.

Architecture

Input Layer



LSTM Layer



Dropout Layer



LSTM Layer



Dense Layer



Forecast Output

Forecast Horizon

- Next 7 Days

- Next 30 Days
 - Next 90 Days
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Key Insights

Insight 1

Cargo demand is strongly influenced by aircraft capacity.

Insight 2

Higher cargo bookings generate higher revenue.

Insight 3

Certain routes consistently outperform others.

Insight 4

Operational delays impact cargo movement efficiency.

Insight 5

Demand forecasting improves capacity planning and utilization.

Business Recommendations

Capacity Optimization

Deploy larger aircraft on high-demand routes.

Demand Forecasting

Integrate predictive models into cargo planning systems.

Delay Reduction

Improve operational efficiency to reduce offloading.

Dynamic Planning

Adjust cargo capacity based on forecasted demand.

Data-Driven Operations

Use analytics to improve route planning and revenue management.

Repository Structure

Airline-Cargo-Demand-Forecasting

```
|
|
|└─ data
|   └─ cargo_operations_5000.csv
|
|└─ notebooks
|   └─ 01_EDA_Cargo_Analysis.ipynb
|   └─ 02_ML_Cargo_Demand_Prediction.ipynb
|   └─ 03_DL_LSTM_Demand_Forecasting.ipynb
|
|└─ reports
|   └─ Final_Report.pdf
|
|└─ README.md
```

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Conclusion

This project demonstrates the application of Data Analytics, Machine Learning, and Deep Learning techniques in airline cargo operations. The developed forecasting framework can help airlines improve demand planning, optimize capacity allocation, reduce offloading, and maximize revenue generation through data-driven decision-making.